For every level the type of n-types is a mere proposition

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 190-192 and its proof environment opens on line 193, the next line; the pairing is by adjacency and the retained proof range 193-206 is the whole environment.

Editorial note. The retained counter values (chapter 7, section 1, theorem offset 9) make the counted theorem print as Theorem 7.1.10, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:isaprop-isofhlevel = 7.1.10).

Editorial note. Three labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:isprop-iscontr (book Lemma 3.11.4, defined at line 991 of the logic chapter pinned file), thm:isprop-forall (book Example 3.6.2, defined at line 575 of the same file) and thm:hlevel-prod (book Theorem 7.1.9, defined at line 171 of this chapter pinned file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the level predicate is a mere proposition at every level (author label thm:isaprop-isofhlevel, book Theorem 7.1.10).

Allowed standard prerequisites (editorial).

- The level predicate. is-type n of X is the type expressing that X is an n-type, defined by recursion on the level with the contractibility predicate at the bottom.. It is the predicate whose propositionality is proved.
- Level of a product of propositions. A product of mere propositions is a mere proposition, and a level predicate over a path type is an instance of that.. It reduces the inductive step to a pointwise statement.
- Contractibility is a proposition. For every type X the type is-contr(X) is a mere proposition.. It is the base case of the induction.

Source reference. Original lemma 3.11.4: sources/hott-logic/pinned-original-source.tex, source lines 991–991. Book Lemma 3.11.4, that the contractibility predicate is a mere proposition, cited as the base case of the induction at source

line 196. Navigation pointer only: the retained proof states the step it needs and is complete.

Source reference. Original remark 3.6.2: sources/hott-logic/pinned-original-source.tex, source lines 575–575. The book Example 3.6.2, that a product of mere propositions is a mere proposition, cited at source line 203 to reduce the product to its factors. Navigation pointer only.

Source reference. Original theorem 7.1.9: sources/hott-hlevels/pinned-original-source.tex, source lines 171–171. Book Theorem 7.1.9, the level of a product of types, cited at source line 203 as the alternative tool for the same reduction. Navigation pointer only: it is counted by its own card of this lane and nothing of the counted proof is delegated to it.

Theorem 7.1.10. *For any $n \geq -2$ and any type X, the type* $\mathsf{is}\text{-}n\text{-}\mathsf{type}(X)$ *is a mere proposition.*

Proof. We proceed by induction with respect to n.

For the base case, we need to show that for any X, the type $\mathsf{isContr}(X)$ is a mere proposition. This is original lemma 3.11.4.

For the inductive step we need to show

$$\prod_{X:\mathcal{U}} \mathsf{isProp}(\mathsf{is}\text{-}n\text{-}\mathsf{type}(X)) \to \prod_{X:\mathcal{U}} \mathsf{isProp}(\mathsf{is}\text{-}(n+1)\text{-}\mathsf{type}(X)).$$

To show the conclusion of this implication, we need to show that for any type X, the type

$$\prod_{x,x':X} \mathsf{is}\text{-}n\text{-}\mathsf{type}(x = x')$$

is a mere proposition. By original remark 3.6.2 or original theorem 7.1.9, it suffices to show that for any $x, x' : X$, the type $\mathsf{is}\text{-}n\text{-}\mathsf{type}(x =_X x')$ is a mere proposition. But this follows from the inductive hypothesis applied to the type $(x =_X x')$. □